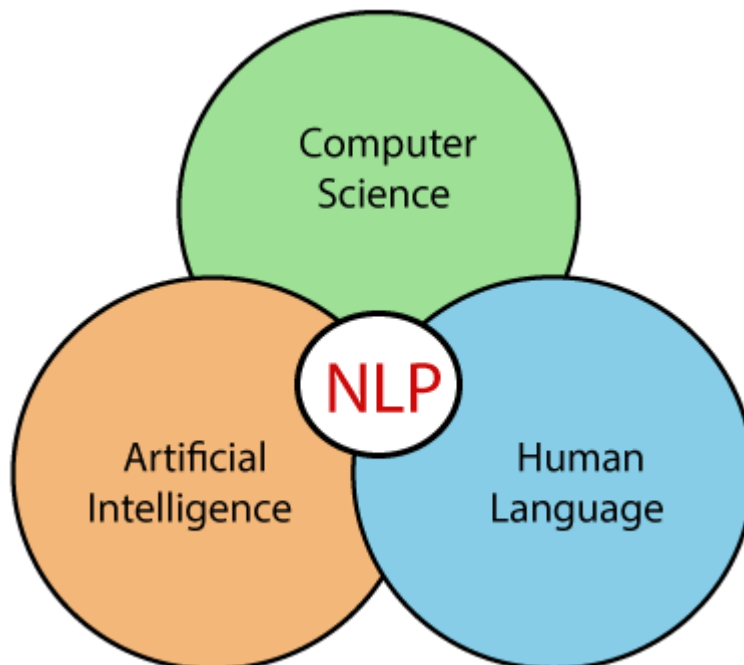


Unit 8: Natural Language Processing (NLP)

Natural language processing (NLP): Natural language processing (NLP) is a branch of artificial intelligence that helps computers understand, interpret and manipulate human language. It is an interdisciplinary subfield of computer science and linguistics. It uses computational linguistics, which is the study of how language works, and various models based on statistics, machine learning, and deep learning. NLP enables computers to comprehend, generate, and manipulate natural language text or voice.

OR

NLP stands for Natural Language Processing, which is a part of Computer Science, Human language, and Artificial Intelligence. It is the technology that is used by machines to understand, analyze, manipulate, and interpret human's languages. It helps developers to organize knowledge for performing tasks such as translation, automatic summarization, Named Entity Recognition (NER), speech recognition, relationship extraction, and topic segmentation.



Applications of NLP

There are the following applications of NLP -

1. Question Answering:

Question Answering focuses on building systems that automatically answer the questions asked by humans in a natural language.

2. Spam Detection:

Spam detection is used to detect unwanted e-mails getting to a user's inbox.

3. Sentiment Analysis:

Sentiment Analysis is also known as opinion mining. It is used on the web to analyze the attitude, behaviour, and emotional state of the sender. This application is implemented through a combination of NLP (Natural Language Processing) and statistics by assigning the values to the text (positive, negative, or neutral), identify the mood of the context (happy, sad, angry, etc.)

4. Machine Translation:

Machine translation is used to translate text or speech from one natural language to another natural language.

5. Spelling correction:

Microsoft Corporation provides word processor software like MS-word, PowerPoint for the spelling correction.

6. Speech Recognition:

Speech recognition is used for converting spoken words into text. It is used in applications, such as mobile, home automation, video recovery, dictating to Microsoft Word, voice biometrics, voice user interface, and so on.

7. Chatbot:

Implementing the Chatbot is one of the important applications of NLP. It is used by many companies to provide the customer's chat services.

8. Information extraction:

Information extraction is one of the most important applications of NLP. It is used for extracting structured information from unstructured or semi-structured machine-readable documents.

9. Natural Language Understanding (NLU)

It converts a large set of text into more formal representations such as first-order logic structures that are easier for the computer programs to manipulate notations of the natural language processing.

Levels of Analysis: In Natural Language Processing (NLP), there are five levels of analysis:

- **Lexical or Morphological Analysis:** Deals with individual words.
- **Syntax Analysis:** Ensures correct sentence structure.
- **Semantic Analysis:** Looks for meaning in a statement.
- **Discourse:** Considers the effect of previous sentences on the current one.
- **Pragmatics:** Focuses on context and language use.

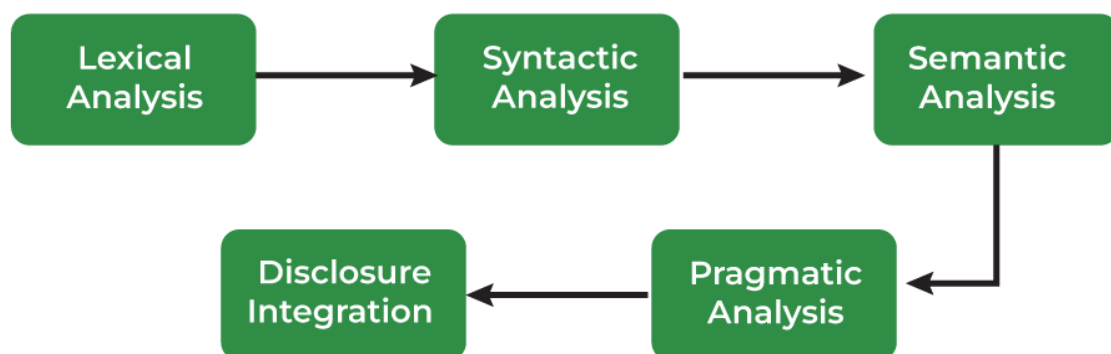


Fig: Phases of Natural Language Processing (NLP)

Tokenization: The lexical phase in Natural Language Processing (NLP) involves scanning text and breaking it down into smaller units such as paragraphs, sentences, and words. This process, known as tokenization, converts raw text into manageable units called tokens or lexemes. Tokenization is essential for understanding and processing text at the word level.

In addition to tokenization, various data cleaning and feature extraction techniques are applied, including:

Lemmatization: Reducing words to their base or root form.

Stop words Removal: Eliminating common words that do not carry significant meaning, such as “and,” “the,” and “is.”

Correcting Misspelled Words: Ensuring the text is free of spelling errors to maintain accuracy.

These steps enhance the comprehensibility of the text, making it easier to analyze and process.

Morphological Analysis

Morphological analysis is another critical phase in NLP, focusing on identifying morphemes, the smallest units of a word that carry meaning and cannot be further divided. Understanding morphemes is vital for grasping the structure of words and their relationships.

Types of Morphemes

Free Morphemes: Text elements that carry meaning independently and make sense on their own. For example, “bat” is a free morpheme.

Bound Morphemes: Elements that must be attached to free morphemes to convey meaning, as they cannot stand alone. For instance, the suffix “-ing” is a bound morpheme, needing to be attached to a free morpheme like “run” to form “running.”

Importance of Morphological Analysis

Morphological analysis is crucial in NLP for several reasons:

- **Understanding Word Structure:** It helps in deciphering the composition of complex words.
- **Predicting Word Forms:** It aids in anticipating different forms of a word based on its morphemes.
- **Improving Accuracy:** It enhances the accuracy of tasks such as part-of-speech tagging, syntactic parsing, and machine translation.

Syntactic Analysis (Parsing)

Syntactic analysis, also known as parsing, is the second phase of Natural Language Processing (NLP). This phase is essential for understanding the structure of a sentence and assessing its grammatical correctness. It involves analyzing the relationships between words and ensuring their logical consistency by comparing their arrangement against standard grammatical rules.

Role of Parsing

Parsing examines the grammatical structure and relationships within a given text. It assigns Parts-Of-Speech (POS) tags to each word, categorizing them as nouns, verbs, adverbs, etc. This tagging is crucial for understanding how words relate to each other syntactically and helps in avoiding ambiguity. Ambiguity arises when a text can be interpreted in multiple ways due to words having various meanings. For example, the word “book” can be a noun (a physical book) or a verb (the action of booking something), depending on the sentence context.

Examples of Syntax

Consider the following sentences:

Correct Syntax: “John eats an apple.”

Incorrect Syntax: “Apple eats John an.”

Despite using the same words, only the first sentence is grammatically correct and makes sense. The correct arrangement of words according to grammatical rules is what makes the sentence meaningful.

Importance of Syntactic Analysis

By analyzing and ensuring proper syntax, NLP systems can better understand and generate human language. This analysis helps in various applications, such as machine translation, sentiment analysis, and information retrieval, by providing a clear structure and reducing ambiguity.

Semantic Analysis

Semantic Analysis is the third phase of Natural Language Processing (NLP), focusing on extracting the meaning from text. Unlike syntactic analysis, which deals with grammatical structure, semantic analysis is concerned with the literal and contextual meaning of words, phrases, and sentences.

Semantic analysis aims to understand the dictionary definitions of words and their usage in context. It determines whether the arrangement of words in a sentence makes logical sense. This phase helps in finding context and logic by ensuring the semantic coherence of sentences.

Key Tasks in Semantic Analysis

Named Entity Recognition (NER): NER identifies and classifies entities within the text, such as names of people, places, and organizations. These entities belong to predefined categories and are crucial for understanding the text’s content.

Word Sense Disambiguation (WSD): WSD determines the correct meaning of ambiguous words based on context. For example, the word “bank” can refer to a financial institution or the side of a river. WSD uses contextual clues to assign the appropriate meaning.

Examples of Semantic Analysis

Consider the following examples:

Syntactically Correct but Semantically Incorrect: “Apple eats a John.”

This sentence is grammatically correct but does not make sense semantically. An apple cannot eat a person, highlighting the importance of semantic analysis in ensuring logical coherence.

Literal Interpretation: “What time is it?”

This phrase is interpreted literally as someone asking for the current time, demonstrating how semantic analysis helps in understanding the intended meaning.

Importance of Semantic Analysis

Semantic analysis is essential for various NLP applications, including machine translation, information retrieval, and question answering. By ensuring that sentences are not only grammatically correct but also meaningful, semantic analysis enhances the accuracy and relevance of NLP systems.

Discourse Integration

Discourse Integration is the fourth phase of Natural Language Processing (NLP). This phase deals with comprehending the relationship between the current sentence and earlier sentences or the larger context. Discourse integration is crucial for contextualizing text and understanding the overall message conveyed.

Role of Discourse Integration

Discourse integration examines how words, phrases, and sentences relate to each other within a larger context. It assesses the impact a word or sentence has on the structure of a text and how the combination of sentences affects the overall meaning. This phase helps in understanding implicit references and the flow of information across sentences.

Importance of Contextualization

In conversations and texts, words and sentences often depend on preceding or following sentences for their meaning. Understanding the context behind these words and sentences is essential to accurately interpret their meaning.

Example of Discourse Integration

Consider the following examples:

Contextual Reference: “This is unfair!”

To understand what “this” refers to, we need to examine the preceding or following sentences. Without context, the statement’s meaning remains unclear.

Anaphora Resolution: “Taylor went to the store to buy some groceries. She realized she forgot her wallet.” In this example, the pronoun “she” refers back to “Taylor” in the first sentence. Understanding that “Taylor” is the antecedent of “she” is crucial for grasping the sentence’s meaning.

Application of Discourse Integration

Discourse integration is vital for various NLP applications, such as machine translation, sentiment analysis, and conversational agents. By understanding the relationships and context within texts, NLP systems can provide more accurate and coherent responses.

Pragmatic Analysis

Pragmatic Analysis is the fifth and final phase of Natural Language Processing (NLP), focusing on interpreting the inferred meaning of a text beyond its literal content. Human language is often complex and layered with underlying assumptions, implications, and intentions that go beyond straightforward interpretation. This phase aims to grasp these deeper meanings in communication.

Role of Pragmatic Analysis

Pragmatic analysis goes beyond the literal meanings examined in semantic analysis, aiming to understand what the writer or speaker truly intends to convey. In natural

language, words and phrases can carry different meanings depending on context, tone, and the situation in which they are used.

Importance of Understanding Intentions

In human communication, people often do not say exactly what they mean. For instance, the word “Hello” can have various interpretations depending on the tone and context in which it is spoken. It could be a simple greeting, an expression of surprise, or even a signal of anger. Thus, understanding the intended meaning behind words and sentences is crucial.

Examples of Pragmatic Analysis

Consider the following examples:

Contextual Greeting: “Hello! What time is it?”

“Hello!” is more than just a greeting; it serves to establish contact.

“What time is it?” might be a straightforward request for the current time, but it could also imply concern about being late.

Figurative Expression: “I’m falling for you.”

The word “falling” literally means collapsing, but in this context, it means the speaker is expressing love for someone.

Application of Pragmatic Analysis

Pragmatic analysis is essential for applications like sentiment analysis, conversational AI, and advanced dialogue systems. By interpreting the deeper, inferred meanings of texts, NLP systems can understand human emotions, intentions, and subtleties in communication, leading to more accurate and human-like interactions.

Difference between Natural language and Computer Language

Natural Language	Computer Language
Natural language has a very large vocabulary.	Computer language has a very limited vocabulary.
Natural language is easily understood by humans.	Computer language is easily understood by the machines.
Natural language is ambiguous in nature.	Computer language is unambiguous.

Parsing and Parse Tree Generation

Parsing: Parsing, syntax analysis, or syntactic analysis is the process of examining a string of symbols that adhere to the rules of a formal grammar, whether they be in natural language, computer languages, or data structures. Parsing is performed at the syntax analysis phase where a stream of tokens is taken as input from the lexical analyzer and the parser produces the parser tree for the tokens while checking the stream of tokens against the syntax errors.

Types of Parsing

The parsing is divided into two types, which are as follows:

1. Top-down Parsing: Top-down parsing attempts to build the parse tree from the root node to the leaf node. The top-down parser will start from the start symbol and proceed to the string. It follows the leftmost derivation.
2. Bottom-up Parsing: Bottom-up parsing builds the parse tree from the leaf node to the root node. The bottom-up parsing will reduce the input string to the start symbol. It traces the rightmost derivation of the string in reverse.

Derivations: Derivations mean replacing a given string's non-terminal by the right-hand side of the production rule. The sequence of applications of rules that makes the completed string of terminals from the starting symbol is known as derivation. The parse tree is the pictorial representation of derivations. Therefore, it is also known as derivation trees. The derivation tree is independent of the other in which productions are used.

A parse tree is an ordered tree in which nodes are labeled with the left side of the productions and in which the children of a node define its equivalent right parse tree also known as syntax tree, generation tree, or production tree.

A Parse Tree for a CFG $G = (V, \Sigma, P, S)$ is a tree satisfying the following conditions:
Root has the label S , where S is the start symbol.

Each vertex of the parse tree has a label which can be a variable (V), terminal (Σ), or ϵ .

If $A \rightarrow C_1, C_2, \dots, C_n$ is a production, then $C_1, C_2, \dots, C_n$ are children of node labeled A .

Leaf Nodes are terminal (Σ), and Interior nodes are variable (V).

The label of an internal vertex is always a variable.

If a vertex A has k children with labels $A_1, A_2, \dots, A_k$, then $A \rightarrow$

Example1 – If CFG has productions.

$S \rightarrow a A S \mid a$
 $S \rightarrow S b A \mid S S \mid b a$

Show that $S \Rightarrow^* aa\ bb\ aa$ & construct parse tree whose yield is $aa\ bb\ aa$.

Solution

$S \Rightarrow^{\text{Im}} \text{Im } a \underline{A} S$

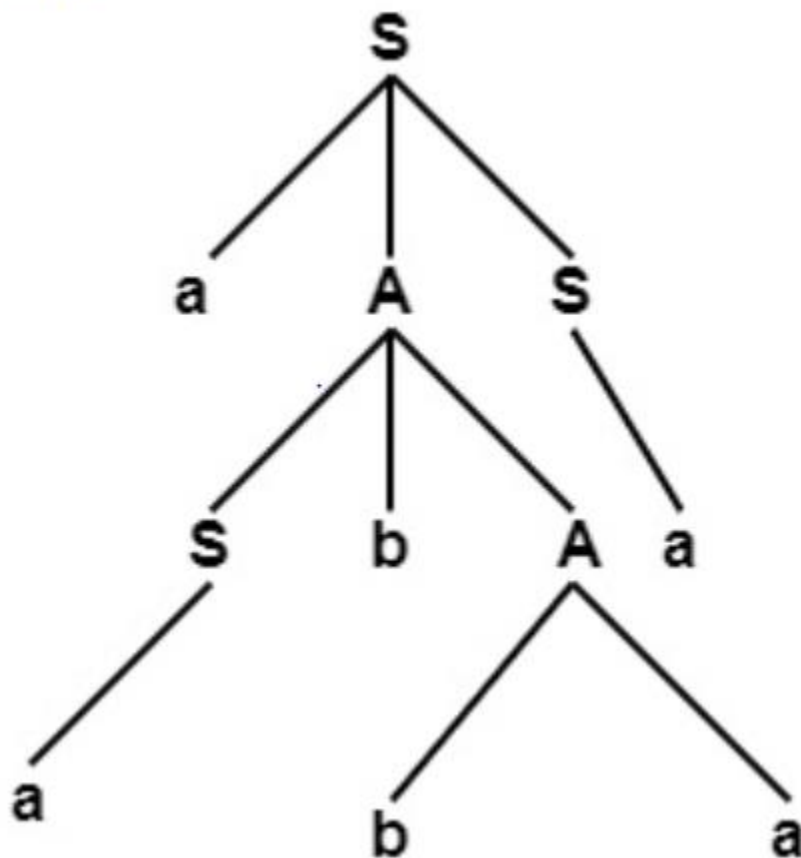
$\Rightarrow a \underline{S} b A S$

$\Rightarrow aa\ b \underline{A} S$

$\Rightarrow aa\ bba \underline{S}$

$\therefore S \Rightarrow^* aa\ bb\ aa$

Derivation Tree



Yield = Left to Right Ordering of Leaves = aa bb aa

Example2: Consider the CFG

$S \rightarrow bB \mid aA$

$A \rightarrow b \mid bS \mid aAA$

$B \rightarrow a \mid aS \mid bBB$

Find (a) Leftmost, Rightmost Derivation for string b aa baba. Also, find derivation Trees.

Solution

Leftmost Derivation

$S \Rightarrow b B$

$\Rightarrow bb BB \quad B \rightarrow bBB$

$\Rightarrow bbaB \quad B \rightarrow a$

$\Rightarrow bbaaS \quad B \rightarrow aS$

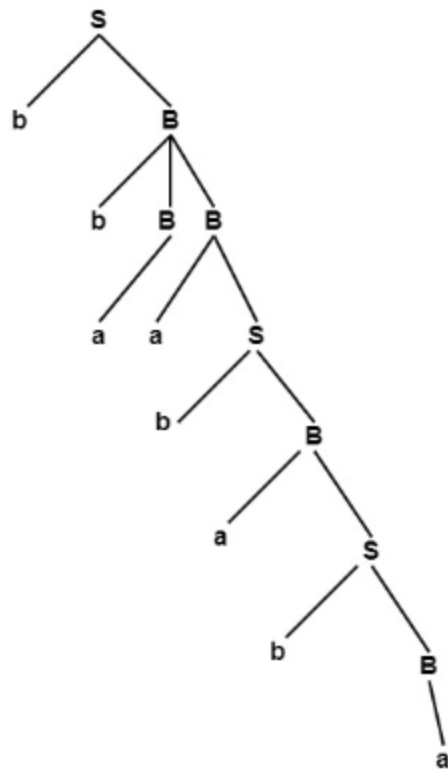
$\Rightarrow bb aabB \quad S \rightarrow bB$

$\Rightarrow bb aa b aS \quad B \rightarrow aS$

$\Rightarrow bb aa bab B \quad S \rightarrow bB$

$\Rightarrow bb aa ba ba \quad B \rightarrow a$

Derivation Tree for Leftmost Derivation



Rightmost Derivation

$S \Rightarrow bB$

$\Rightarrow bb BB$

$\Rightarrow bbBaS$

$\Rightarrow bbBabB$

$\Rightarrow bbBabaS$

$\Rightarrow bbBababB$

$\Rightarrow bbBababa$

$\Rightarrow bbaababa$

Derivation Tree for Rightmost Derivation

